Cleaning, Reshaping and Analyzing Sales Data in Power BI: My Practical Experience

For this practical, I went back into the Adventure Works dataset in Power BI, but this time the focus was less about basic math and more about reshaping data and actually building some visuals from it. If you're new to Power BI, don't worry, I'll explain things simply as I go.

Quick refresher: Power Query is the part of Power BI where you clean and reshape your data before you build anything with it. Think of it as prepping ingredients before you cook. This practical had two main parts, Power Query operations and then actual sales analysis, plus a few advanced tasks at the end.

Part A: Reshaping the Data in Power Query

This part had four smaller sections, each teaching a different way to reshape data.

Grouping. This is basically summarizing data. Instead of looking at every single sales row, I grouped rows by things like product category or customer, and let Power Query add up, average or count the values for me. For example, grouping by product category and summing the sales amount gives you one total per category instead of thousands of individual rows.

Pivoting and unpivoting. This one confused me at first. Unpivoting takes columns like Sales Amount, Tax Amount and Order Quantity, which are normally side by side, and stacks them into two columns instead, one saying which metric it is and one saying the value. Pivoting does the opposite, it takes that stacked format and spreads it back into separate columns. I did both just to see the data go one way and then come back to how it started, which also helped me check that nothing got lost in the process.

Append queries. This is just stacking two tables on top of each other. I split the sales data into a first half and a second half based on the order date, then appended them back into one table. It's the same idea as combining two Excel sheets with the same columns into one.

Merge queries. This is closer to what you'd call a join in databases or VLOOKUP in Excel. My sales table only had product ID numbers, not actual product names or categories, so I merged in extra tables to pull in the product name, its subcategory and its category, step by step, until every sales row had the full product info attached.

Part B: Turning the Data into Visuals

Once the data was cleaned and merged, this part moved into the actual Report view of Power BI, where you build charts.

I made simple bar charts comparing total sales by product category, by sales territory, and separate ones showing the top 10 products and top 10 customers by sales. I also built a couple of Card visuals, which just show one big number, to instantly display the highest selling product and the top revenue territory without needing to scroll through a whole chart. Other charts covered sales by year and by month, and average tax collected per territory. Nothing

too complicated here, mostly just picking the right chart type and dragging the right fields onto it.

Advanced Tasks: Going a Bit Deeper

The last section pushed a bit further using tools still inside Power Query. I ranked product categories by total sales using an index column after sorting them, flagged which products were selling above the overall average, grouped customers into high, medium and low spenders based on how much they bought, and calculated what percentage each product category contributed to total sales. I also compared weekend versus weekday sales by pulling out the day name from the order date, and created sales bands that label each transaction as low, medium or high value.

What I took away from this

This practical made it clearer to me that Power BI work is less about clicking a chart button and more about getting your data into the right shape first. Grouping, merging and reshaping data properly makes the actual visualizing part almost effortless afterward. If you're starting out, I'd say don't rush past the Power Query steps just to get to the charts, that's really where most of the real work happens.